XOR and breaking linear models

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2026-02-23

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The XOR problem and the defeat of linear models

In this exercise, we will use the XOR function to motivate nonlinear elements in linear models, then learned-parameter linear models with nonlinear elements. The goal is to find generalizable models that can solve a classifier question: given 2D binary inputs, can we solve XOR?

Introduction to XOR

XOR is a function which takes two binary inputs, returning 1 for OR inputs and 0 for all others. It can be described as $\mathrm{XOR} : \{0, 1\} \to \{0, 1\}$ and as a table with:

x_1	x_2	$\mathrm{XOR}(x_1, x_2)$
0	0	0
0	1	1
1	0	1
1	1	0

Linear model approach

A simple question might be to predict the outputs with a useful model, such that inputing $x = \{x_1, x_2\}$ would correctly model XOR with an output $\hat{y}$. A naïve approach might be to use a linear model.

$$\hat{y} = w_1 x_1 + w_2 x_2 + b = w^\top x + b \tag{1}$$

We can solve for each input and output of XOR.

$$\begin{aligned} x_1 = 0, x_2 = 0, y = 0, &\quad 0 = w_1(0) + w_2(0) + b, &\quad b = 0 \\ x_1 = 0, x_2 = 1, y = 1, &\quad 1 = w_1(0) + w_2(1) + b, &\quad b = 1 - w_2 \\ x_1 = 1, x_2 = 0, y = 1, &\quad 1 = w_1(1) + w_2(0) + b, &\quad b = 1 - w_1 \\ x_1 = 1, x_2 = 1, y = 0, &\quad 0 = w_1(1) + w_2(1) + b, &\quad b = -w_1 - w_2 \end{aligned} \tag{2}$$

This is a contradiction. Plugging in $b = 0$, $w_1 = w_2 = 1$, which contradicts with $b = -w_1 - w_2$. Thus, this model cannot solve XOR.

This contradiction can be formalized with a more rigorous proof:

Theorem 1 (Proof by contradiction: Two dimensional linear models will not solve XOR.).

Proof. A general affine two dimensional model that maps $\mathbb{R}^2 \to \mathbb{R}$ can be defined as:

$$f(x_1, x_2) = Ax_1 + Bx_2 + C \tag{3}$$

In order for Equation 3 to solve XOR $f(0,0) + f(1,1) \neq f(0,1) + f(1,0)$.

$$\begin{aligned} f(0,0) + f(1,1) &= A(0) + B(0) + C + A(1) + B(1) + C \\ &= A + B + 2C \\ f(0,1) + f(1,0) &= A(0) + B(1) + C + A(1) + B(0) + C \\ &= A + B + 2C \end{aligned}$$

Thus, in a linear model, $f(0,0) + f(1,1) = f(0,1) + f(1,0)$, which contradicts the XOR function.

■

Can a deeper linear model solve XOR?

Suppose more terms are added to the original linear model Equation 1, as such:

$$\begin{aligned} \hat{y} &= w_1 h_1 + w_2 h_2 + b \\ h_1 = a_1 x_1 + a_2 x_2 + c_1, &\quad h_2 = a_3 x_1 + a_4 x_2 + c_2 \end{aligned} \tag{4}$$

This can be viewed as a nested linear model, where linear terms are themselves composed of linear terms:

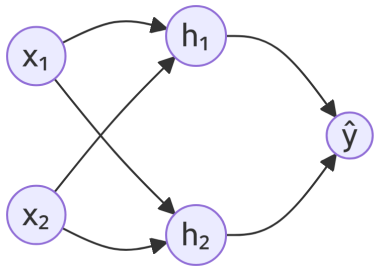


Figure 1: Architecture of the nested linear model.

A nested linear model can still be simplified to a similar form to the original linear model:

$$\begin{aligned}\hat{y} &= h_1(a_1x_1 + a_2x_2 + c_1) + h_2(a_3x_1 + a_4x_2 + c_2) + b \\ &= h_1a_1x_1 + h_1a_2x_2 + h_1c_1 + h_2a_3x_1 + h_2a_4x_2 + h_2c_2 + b \\ &= h_1a_1x_1 + h_2a_3x_1 + h_1a_2x_2 + h_2a_4x_2 + h_1c_1 + h_2c_2 + b \\ &= (h_1a_1 + h_2a_3)x_1 + (h_1a_2 + h_2a_4)x_2 + h_1c_1 + h_2c_2 + b\end{aligned} \tag{5}$$

The overall model reduces to a single linear combination:

$$\begin{aligned}\hat{y} &= Ax_1 + Bx_2 + C \\ A = h_1a_1 + h_2a_3, &\quad B = h_1a_2 + h_2a_4\end{aligned}$$

Since this model reduces to the same two dimensional affine model Equation 3, it too will follow the same contradiction in Equation 2. So, a deeper linear model will not solve XOR.

Can more terms in a linear model solve XOR?

Our proof Theorem 1 showed that a linear model with 2 terms, equal to the input, does not solve XOR. So what if there were more terms? Note, since the input is 2 dimensional, a third term must be some linear combination of the terms.

$$\begin{aligned}\hat{y} &= Ax_1 + Bx_2 + Dx_3 + C \\ x_3 &= px_1 + qx_2 + r\end{aligned} \tag{6}$$

Expanding this model:

$$\begin{aligned}\hat{y} &= Ax_1 + Bx_2 + D(px_1 + qx_2 + r) + C \\ &= (A + Dp)x_1 + (B + Dq)x_2 + Dr + C\end{aligned}$$

This once again follows the form of Equation 3, and follows the contradiction of Theorem 1.

This is also a good opportunity to look at another (more geometric) proof for why linear combinations alone can't solve XOR.

Theorem 2 (Proof by midpoint: n-dimensional linear models will not solve XOR with affine embeddings.). *Here is a proof that generalizes to any linear model, regardless of depth (which is not explicitly stated, but inferred from the above work) and mapping $\mathbb{R}^n \to \mathbb{R}$. It assumes that inputs are affine only.*

Proof. For any affine function $f : \mathbb{R}^n \to \mathbb{R}$,

$$\hat{y} = w^\top x + b \tag{7}$$

Suppose a function (e.g., a line) separates two classes of data if coefficients exist such that $f(x) > 0$ for class 1 and $f(x) < 0$ for class 2.

Assume points P and Q are in class 1, and their midpoint is $M_{P,Q} = \frac{P+Q}{2}$.

The value of an affine function f at the midpoint M is equal to the average of its values at the endpoints P and Q if $f(M) = \frac{f(P)+f(Q)}{2}$.

$$\begin{aligned} f(M_{P,Q}) &= w^\top M + D = w^\top \left(\frac{P+Q}{2} \right) + D \\ &= \frac{w^\top P + w^\top Q}{2} + D \end{aligned}$$

$$\begin{aligned} f(\text{average}_{P,Q}) &= \frac{f(P) + f(Q)}{2} = \frac{\left(w^\top P + D\right) + \left(w^\top Q + D\right)}{2} \\ &= \frac{w^\top P + w^\top Q}{2} + D \end{aligned}$$

Thus, $f(M_{P,Q}) = f(\text{average}_{P,Q})$. Since $f(P) > 0$ and $f(Q) > 0$, $f(M_{P,Q}) > 0$.

Suppose there are points R and S where $f(R) < 0$ and $f(S) < 0$. These points will be in class 2. Following the algebra above, $f(M_{R,S}) < 0$ if f separates class 1 and class 2.

For the XOR function, assign $P = (0,0), Q = (1,1)$ in class 1 (due to equivalent output) and $R = (0,1), S = (1,0)$ in class 2. Then,

$$M_{P,Q} = \frac{\begin{bmatrix} 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 1 \end{bmatrix}}{2} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$M_{R,S} = \frac{\begin{bmatrix} 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \end{bmatrix}}{2} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

Since $M_{P,Q} = M_{R,S}$, there is no f that satisfies $f(M_{P,Q}) > 0$ and $f(M_{R,S}) < 0$ and separates between these classes for XOR. ■

The geometric motivation is that the midpoint geometry is unchanged regardless of linear transformation. An intuitive understanding could be learned from a worked example before the generalized proof.

Beginning with the affine function $\mathbb{R}^3 \to \mathbb{R}$:

$$\begin{aligned} \hat{y} &= Ax_1 + Bx_2 + Dx_3 + C \\ x_3 &= px_1 + qx_2 + r \end{aligned} \tag{8}$$

The transformed points can be notated as z_{x_1,x_2}, where

$z_{0,0} = \begin{bmatrix} 0 \\ 0 \\ r \\ C \end{bmatrix}$ and $z_{0,1} = \begin{bmatrix} 0 \\ 1 \\ q+r \\ C \end{bmatrix}$.

The midpoint between these points can be called m, where m_0 is between the endpoints $z_{0,0}$ and $z_{1,1}$ and m_1 is between the endpoints $z_{0,1}$ and $z_{1,0}$. Since C is a simply the bias term for prediction, it can be ignored for the sake of simplicity. Note that these midpoints are of endpoints decided *a priori* based on how they should classify in the XOR function.

Calculating the midpoints:

$$m_0 = \frac{z_{0,0}, z_{1,1}}{2} = \frac{\begin{bmatrix} 0 \\ 0 \\ r \end{bmatrix} + \begin{bmatrix} A \\ B \\ A+B+p+q+r \end{bmatrix}}{2} = \frac{\begin{bmatrix} A \\ B \\ A+B+p+q+2r \end{bmatrix}}{2}$$

$$m_1 = \frac{z_{0,1}, z_{1,0}}{2} = \frac{\begin{bmatrix} 0 \\ B \\ B+q+r \end{bmatrix} + \begin{bmatrix} A \\ 0 \\ A+p+r \end{bmatrix}}{2} = \frac{\begin{bmatrix} A \\ B \\ A+B+p+q+2r \end{bmatrix}}{2}$$

Thus, $m_0 = m_1$, which won't solve XOR per Theorem 2.

Can a linear model containing a nonlinear term solve XOR?

Changing the model to:

$$\begin{gathered} \hat{y} = Ax_1 + Bx_2 + Dx_3 + C \\ x_3 = x_1 x_2 \end{gathered} \tag{9}$$

Following Theorem 1 with this new model:

$$\begin{aligned} f(0,0) + f(1,1) &= A(0) + B(0) + D(0) + C + A(1) + B(1) + D(1) + C \\ &= A + B + D + 2C \\ f(0,1) + f(1,0) &= A(0) + B(1) + D(0) + C + A(1) + B(0) + D(0) + C \\ &= A + B + 2C \end{aligned}$$

This does not follow the same contradiction. Similarly, the midpoints will not be equivalent since $f(P) + f(Q) \neq f(R) + f(S)$ and there must exist a classifying function f, per Theorem 2.

$$\begin{gathered} f(0,0) = C = 0 \\ f(1,1) = A + B + D + C = 0 \\ f(0,1) = B + C = 1 \quad B = 1 \\ f(1,0) = A + C = 1 \quad A = 1 \\ 1 + 1 + D + 0 = 0 \quad D = -2 \end{gathered}$$

So, $\hat{y} = x_1 + x_2 - 2x_1 x_2$ is a solution to XOR.

It is clear that the model is a linear prediction in nonlinear transformed space with more succint notation:

$$\begin{aligned}\hat{y} &= Ax_1 + Bx_2 + D(x_1x_2) + C \\ &= w^\top \phi(x) + b\end{aligned}$$

where

$$w^\top = [A \;\; B \;\; D] \quad \phi = \begin{bmatrix} x_1 \\ x_2 \\ x_1x_2 \end{bmatrix}$$

Do nonlinear terms need to be declared?

Above, a nonlinear term was guessed and was found to solve XOR. A more generalizable learning function would be to not directly specify this term. For example, applying a generic nonlinear function to the output of the simple affine equation Equation 1.

$$g(w_1x_1 + w_2x_2 + b) \tag{10}$$

where g is any nonlinear function. Suppose $g(z) = z^2$, then

$$\begin{aligned}g(w_1x_1 + w_2x_2 + b) &= (w_1x_1 + w_2x_2 + b)^2 \\ &= (w_1x_1)^2 + (w_2x_2)^2 + b^2 + 2w_1x_1w_2x_2 + 2w_1x_1b + 2w_2x_2b\end{aligned}$$

The same interaction term, x_1x_2, appears here as well. Without solving, this means there will be a solution to XOR if this model is added to Equation 1. For exercise of solving, the model can be simplified to drop the bias term and make the algebra simpler.

$$\begin{gathered}\hat{y} = w_1x_1 + w_2x_2 + w_3(x_1 + x_2)^2 \\ f(0,0) = 0 \quad f(1,1) = w_1 + w_2 + 4w_3 = 1 \\ f(0,1) = w_2 + w_3 = 1 \\ f(1,0) = w_1 + w_3 = 1\end{gathered}$$

Solving as a system of linear equations with row substitution. Rows can be added or multiplied together:

$$\begin{bmatrix} 1 & 1 & 4 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ & & & \end{bmatrix} \overset{R_1 \leftarrow R_1 - R_3}{\rightarrow} \begin{bmatrix} 0 & 1 & 3 & -1 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ & & & \end{bmatrix} \overset{R_1 \leftarrow R_1 - R_2}{\rightarrow} \begin{bmatrix} 0 & 0 & 2 & -2 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ & & & \end{bmatrix}$$

$$w_3 = -1 \quad w_1 = 2 \quad w_2 = 2$$

Showing equivalence to Equation 9:

$$
\begin{aligned}
\hat{y} &= 2x_1 + 2x_2 - (x_1 + x_2)^2 \\
&= 2x_1 + 2x_2 - (x_1^2 + x_2^2 + 2x_1x_2) \\
&= (2x_1 - x_1^2) + (2x_2 - x_2^2) - 2x_1x_2 \\
&\text{since elements in x are binary, } \; x_i^2 = x_i \\
&= x_1 + x_2 - 2x_1x_2
\end{aligned}
$$

What if all the terms of the linear model are nonlinear?

Suppose Equation 9 is updated to be:

$$
\begin{aligned}
\hat{y} &= v_1h_1 + v_2h_2 \\
h_1 &= \sigma(a_1x_1 + b_1x_2 + c_1) \\
h_2 &= \sigma(a_2x_1 + b_2x_2 + c_2)
\end{aligned}
\tag{11}
$$

where σ is a nonlinear function. Continuing the previous example, h can be set to $(x_1 - x_2)^2$, producing:

$$
\hat{y} = v_1(x_1 - x_2)^2 + v_2(x_1 - x_2)^2
$$